

print ("Area of circle", c1)

011 Enter a length of rec : 23

Enter a breadth of rec : 12

Enter a base of tri : 65

Enter a height of tri : 24

Enter a radius of circle : 5

Area of Rectangle : 276.0

Area of Triangle : 500.0

Area of circle : 78.55

* Operators

⇒ It is a special symbol used for particular operation between two operands.

Ex: $a + b = c$ [a, b, c → operands] [+ → operators]

Types

1] Arithmetic operator

→ In this we calculate the addition, subtraction, multiplication, division and modulus, float division and power.

Types

+, -, *, /, %, //, **

Ex:

Read 2 integer numbers from user and perform all the arithmetic operations.

Prog.

```
a = int(input("Enter a number:"))
```

```
b = int(input("Enter a number:"))
```

```
print("The sum of a & b is a+b")
ln The sub of a & b is a-b ln The
mul of a & b is a*b ln The div of
a & b is a/b ln The mod of a & b
is a%b ln The float division of a & b
is a/b ln The power of a to the power of b is a**b
ln")
```

O/P

Enter a number : 12

Enter a number : 4

Sum of 12 & 4 is : 16

Sub of 12 & 4 is : 8

Mul of 12 & 4 is : 48

division of 12 & 4 is : 3.0

mod of 12 & 4 is : 0

Float division of 12 & 4 is : 3

The 12 Power of 4 is 20736

2) Assignment Operators

⇒ It is used to assign a value to a variable.

Types

=, +=, -=, *=, /=, %=, **=

Ex:

A = 23

A += 23

A -= 8

A *= 3

A # 23

A # 68

A # 60

A # 20

3) Logical Operators

⇒ It is used for combine conditional statements & perform Boolean operations.

Types

and, or, not

a	b	and	or	not	xor	xnor	nand
0	0	0	0	1	0	1	1
0	1	0	1	0	1	0	1
1	0	0	1	0	1	0	1
1	1	1	1	0	0	1	0

Ex: age = 24

country = "india"

age >= 18 and country == "india" # True

Q4) Relational operators

⇒ It is used for comparing the two variables.

Types

$=$, $!=$, $>$, $<$, $>=$, $<=$

Ex:	$a = 20$	$a > b$	$a = b$	$a < b$	$a >= b$
	$b = 25$	False	False	True	False

5) Bitwise operators

⇒ It is used to perform operations directly on the individual bits (binary representations of integer operand).

Types

$\&$, $|$, $\sim$, $\gg$, $\ll$, $\wedge$

$2^2 = 4 \rightarrow 0, 1, 2, 3$

$2^3 = 8 \rightarrow 0, 1, 2, 3, \dots, 7$

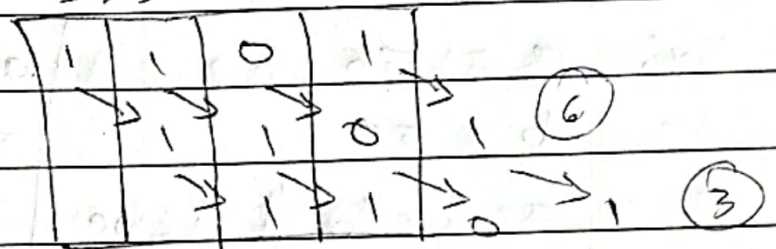
	4	2	1	
	2^2	2^1	2^0	
	0	0	0	0
	0	0	1	1
	0	1	0	2
	0	1	1	3
	1	0	0	4

4	2	1	9
1	0	1	5
1	1	0	6
1	1	1	7

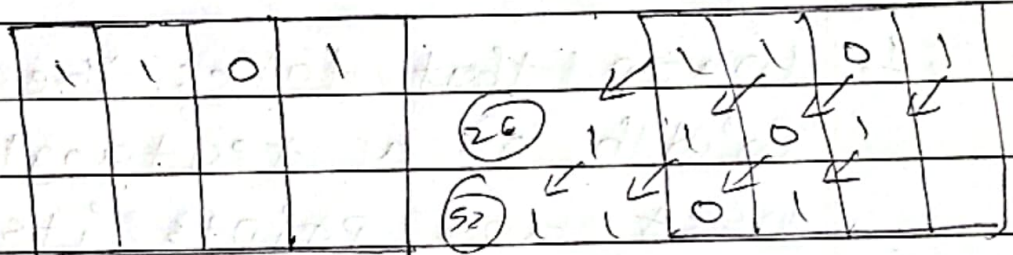
Right

Ex: $12 = 1100$ $9 = 1001$ $8 = 1000$ $1 = 0001$

(13)

 $13 > 2$  $13 < 2$

Left

 $13 < 2$ 

6) Membership operators

⇒ It is used to check if a value or variable is present in the list.

Types in, not in

Ex: $gap = [1, 4, 6, 8, 9, 13]$ 4 in gap

10 in gap

True

False

6 not in gap

False

7) Identity Operators

⇒ It is used to check if two variables are same or not.

Types 'is', 'is not'

Ex:	$a = 56$	$a \text{ is not } b$
	$b = 56$	False
	$a \text{ is } b$	True

HW

1. Wapp that takes the length and width of a rectangle from the user and prints its area.

Prog

```
length = float(input("Enter the length"))
```

```
width = float(input("Enter the width"))
```

```
area = length * width
```

```
print("The Area of Rectangle:", area)
```

O/P

```
Enter the length : 13
```

```
Enter the width : 45
```

```
The Area of Rectangle : 585.0
```


2. Wapp that asks the user for the side of square and prints its perimeter.

prog `side = float(input("Enter a side "))`

`Perimeter = 4 * side`

`print("The perimeter of the square is:",
Perimeter)`

O/P Enter a side : 5
The perimeter of the square is : 20.0

3. Take the base & height of a triangle as input and print its area.

prog `height = float(input("Enter a height"))
base = float(input("Enter a base"))`

`area = 0.5 * height * base`

`print("Area of triangle:", area)`

O/P Enter a height : 12
Enter a base : 34
Area of triangle : 204.0

4. Wapp that asks the user for the radius of a circle & prints its cir

- circumference. (use 3.14 for π)

prog Radius = float(input("Enter a Radius"))

circumference = $2 * 3.14 * \text{Radius}$

print("The circumference of circle is",
circumference)

o/p The Enter a Radius : 23
The circumference of circle : 144.44

5. Take Principal (P), Rate (R), and Time (T) as input from user and print the simple interest.

prog P = float(input("Enter a Principal"))

T = float(input("Enter a Time"))

R = float(input("Enter a Rate"))

SI = $(P * T * R) / 100$

print("The Simple Interest:", SI)

o/p Enter a Principal : 12

Enter a Time : 9

Enter a Rate : 56

The Simple Interest is : 60.48